STATS 608 Final presentation: Bayesian time-series for mobile health

Andrej Leban & Luke Francisco
April 2024

Problem motivation

- We wish to address two common issues in mobile health data (such as smart watch measurements):
  - *Imputation* - missing measurements
  - *Prediction* - either subject dropout OR for real-time performance reasons

- We wish to take advantage of flexible *Bayesian structural time series* methods that account for variation between the subjects

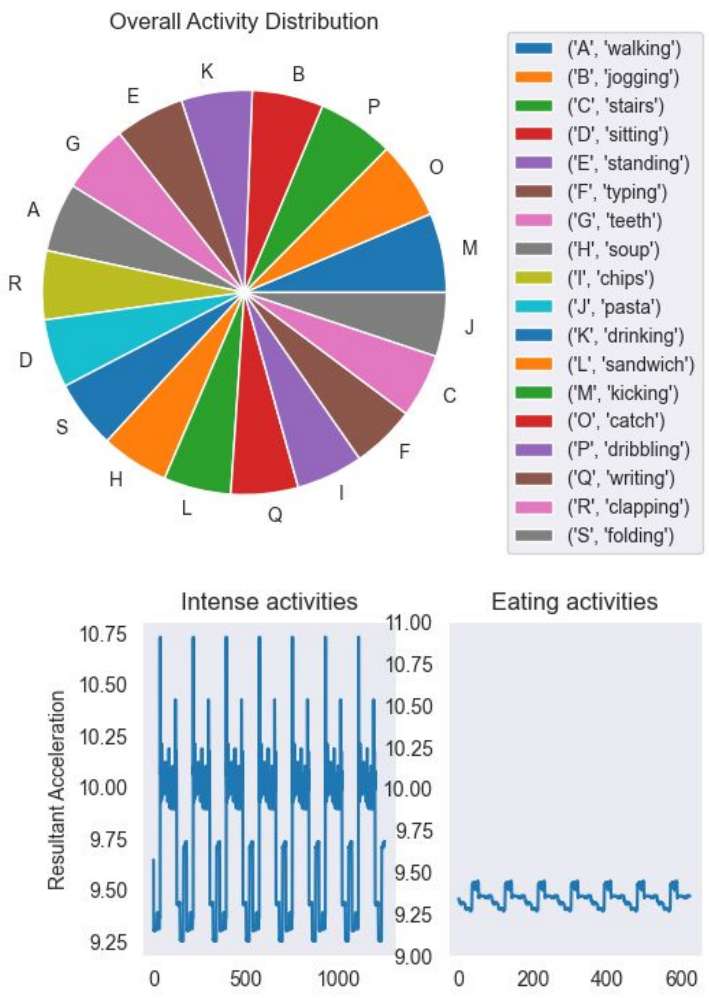
- We evaluate these (potentially) complex models using *Hamiltonian Monte Carlo*, specifically the *No-U-Turn Sampler* implemented in *PyMC*

The data

- *WISDM Smartphone and Smartwatch Activity and Biometrics Dataset*: accelerometer & gyroscope measurements collected with smartphone & watch during 18 different activities across 44 subjects

- Significant differences between different activities & across subjects

- Due to privacy issues, no personal characteristics available



Modeling approach

- For this reason, we derive *static* covariates from the *low-activity* data (such as eating), such as: *overall averages of the time-series measurements, the averages of the peaks, the per-person minutes of activity logged, the entropy of the measurement distributions within a time window*

- These should be a good indicator of a person's *intrinsic* activity level

- We then make use of them to model each *high-activity* time series conditionally on the person and activity performed

- For this project, we focus on the *accelerometer resultant* as our outcome:

$$Res_i = \sqrt{\dot{X}_i^{\,2} + \dot{Y}_i^{\,2} + \dot{Z}_i^{\,2}}$$

The models

- We use *AR(1), AR(2),* and *AR(3)* structural models
- For **prediction**, we use the following model:

$$Y_t = C + C_{subject} + C_{activity} + f(X, \text{subject}, \text{activity}) + \sum_{i=0}^{p} \phi_i Y_{t-i} + \epsilon_t$$

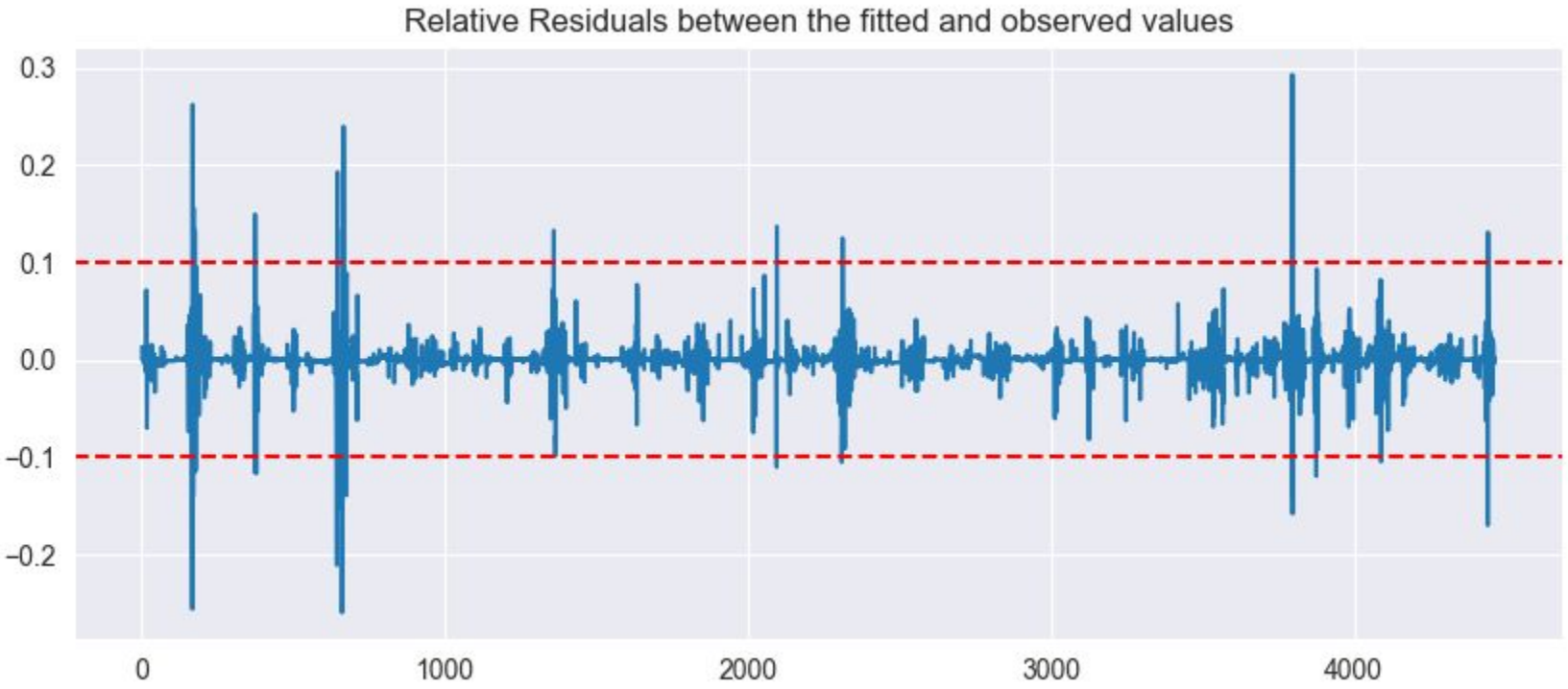
where *X* are the derived static covariates, and *p* is the AR order

- For **imputation**, we also use concurrent measurements of other available quantities *Z*:

$$Y_t = C + C_{subject} + C_{activity} + f(X, \text{subject}, \text{activity}) + \sum_{j} \alpha_j Z_{j,t} + \sum_{i=0}^{p} \phi_i Y_{t-i} + \epsilon_t$$

- We mostly use standard normal / half normal priors

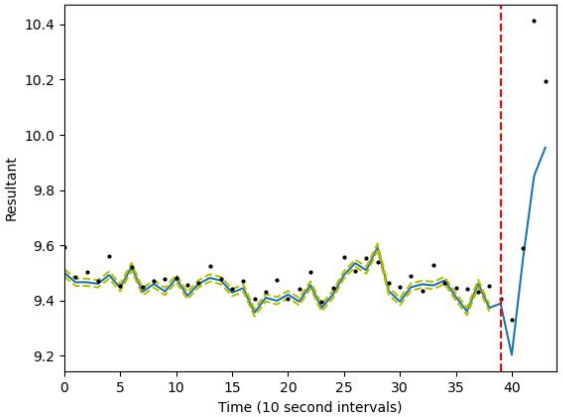
Results - overall fit for AR(1)

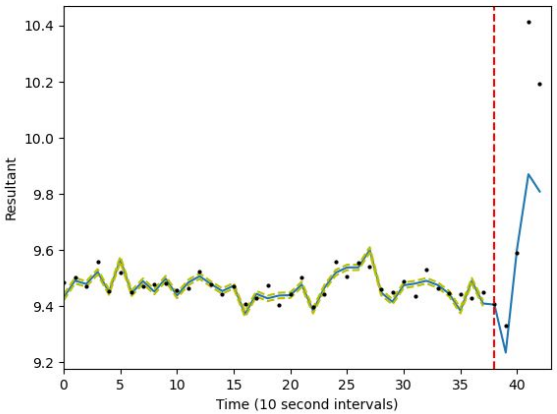


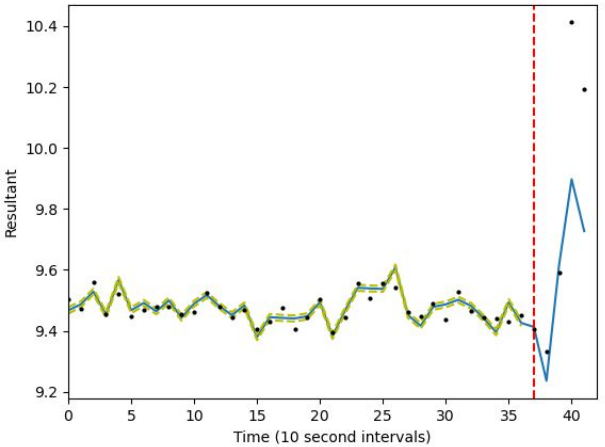
Mostly under 10% relative error!

Results - imputation: AR(1), AR(2), AR(3)

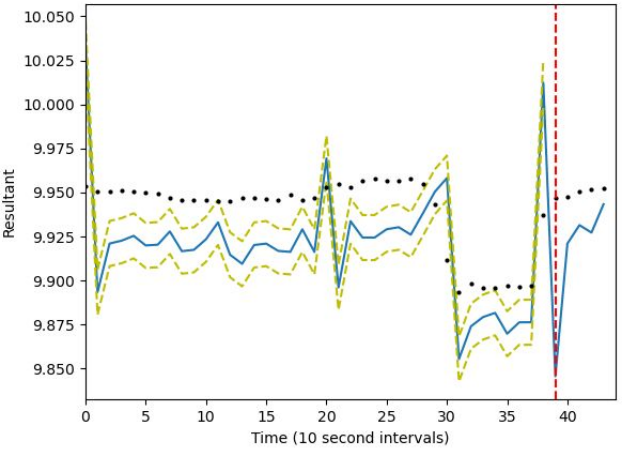
- same subject & activity across all plots. We withhold the last 5 points, but use future values of *other measurements*

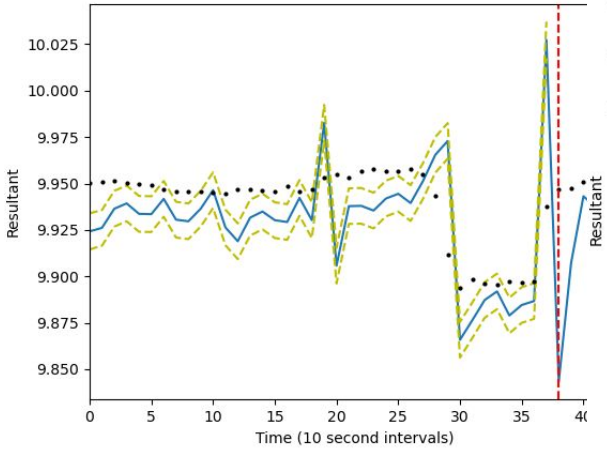


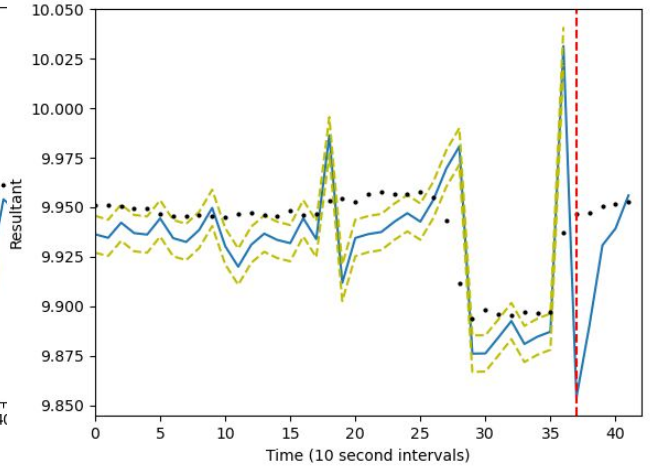




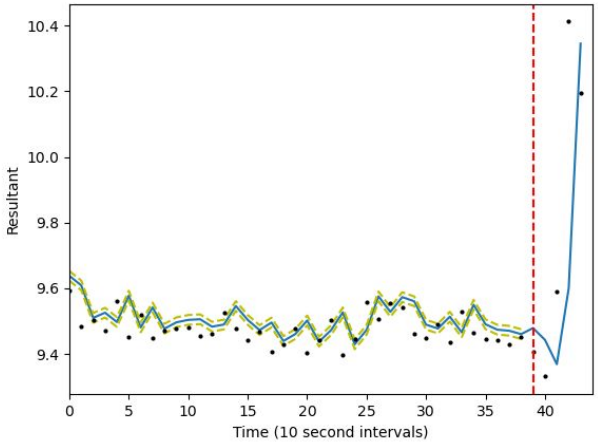
Results - imputation AR(1), AR(2), AR(3) cont'd

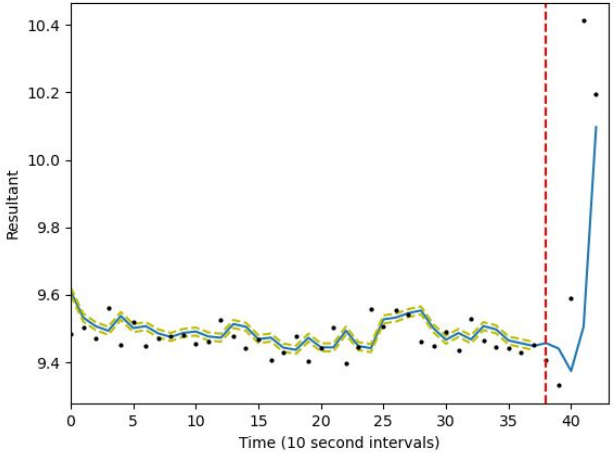


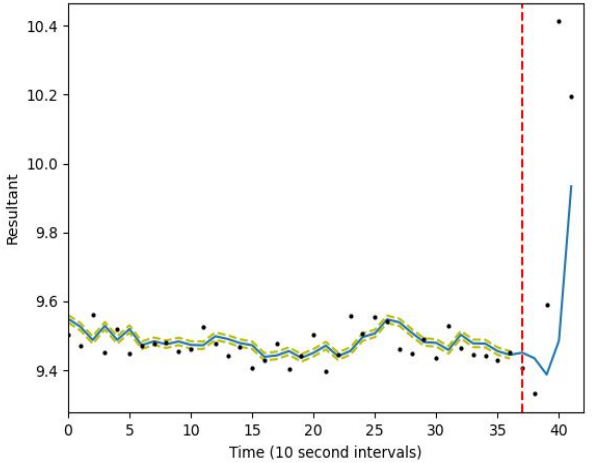




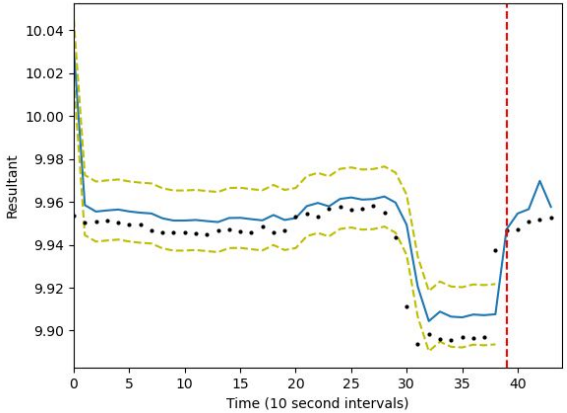
Results - prediction AR(1), AR(2), AR(3)

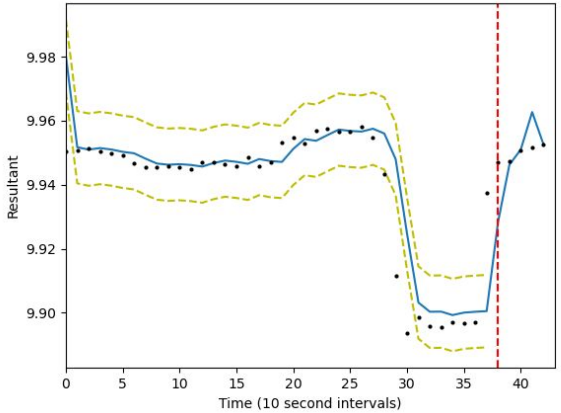


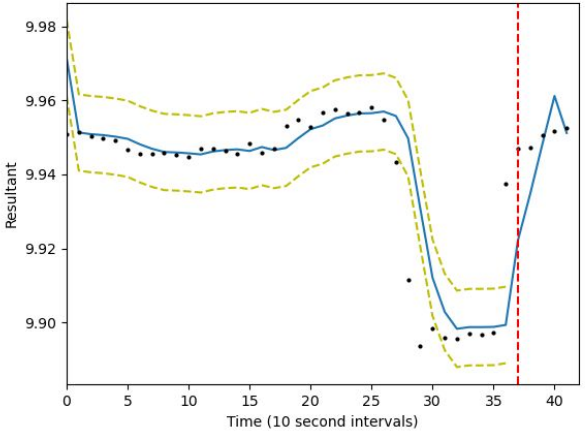




Results - prediction AR(1), AR(2), AR(3)







Conclusions

- Even relatively low-order AR processes, helped by structural factors and a suitable choice of priors, achieve very good “fit” of the observed data

- Likewise, imputation and prediction perform very well even with such relatively simple models

- However, there are indications that the generating process for such data might be more complex than assumed, as hinted at by the “spikes” in activity the models struggled with

- These results seem promising for more involved mobile health data sets where good imputation is currently a stumbling block for downstream tasks